TERMINAL, TELEPHONE AN/TCC-24( )

1. SCOPE

1.1 This specification covers a compact, lightweight, transistorized telephone terminal which provides four channel operation and an order wire for use on loaded spiral-four cable or radio relay facilities or combination of loaded spiral-four cable and radio facilities.

1.2 This specification also covers a four channel hybrid transformer unit used with the transistorized Telephone Terminal AN/TCC-24( ).

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, and drawings of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Boxes, Fiber
PPP-T-60	Tape, Pressure Sensitive Adhesive, Waterproof, for Packaging and Sealing

MILITARY

MIL-P-116	Preservation, Methods of
MIL-B-117	Bags, Interior Packaging
MIL-P-11268	Parts, Materials, and Processes Used in Signal Corps Equipment
MIL-I-11748	Interference Reduction for Electrical and Electronic Equipment
MIL-H-11996	Handset H-60( )/U
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Signal Equipment
MIL-E-15090	Enamel, Equipment, Light-Gray (Formula No. 11)

STANDARDS

FEDERAL

No. 595	Colors

FSC 5805

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MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-169	Extreme-Temperature Cycle
MIL-STD-170	Moisture Resistance Test Cycle for Ground Signal Equipment
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects

DRAWINGS

SIGNAL CORPS

SC-C-33073	Form to be Used for Pilot Pack Process Sheet
SD-DL-72398	Binding Post U-106/U

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer. Both the title and identifying number or symbol should be stipulated when requesting copies.)

2.2 Other publications.- The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

AMERICAN TRUCKING ASSOCIATION

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, 1424 16th Street, N.W., Washinton 6, D. C.)

ASSOCIATION OF AMERICAN RAILROADS

Uniform Freight Classification

(Applications for copies should be addressed to the Association of American Railroads, 202 Chicago Union Station, Chicago 6, Illinois.)

3. REQUIREMENTS

3.1 Description.- Transistorized Telephone Terminal AN/TCC-24( ) is a frequency division multiplexer which provides 4 telephone carrier channels and an order wire. It is capable of operation over up to 40 miles of loaded spiral-4 cable consisting of Cable Assembly CX-1065( )/G and Telephone Loading Coil Assembly CU-260( )/G, and over radio relay facilities.

3.1.1 The 4-Channel hybrid transformer unit is a 4-wire to 2-wire hybrid transformer to permit connection to existing Signal Corps Equipment.

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3.2 Parts, materials, and processes; general.- In addition to any requirements of this specification, covering parts, materials, and processes, such items shall conform to Specification MIL-P-11268 including the selection requirements therein.

3.3 Finish.- The equipment shall be given protective finish in accordance with Specification MIL-F-14072. The color of the transit case shall be No. 2430 of Federal Standard 595. The front panel shall be painted with semi-gloss light-gray enamel conforming to Specification MIL-E-15090.

3.4 Marking.-

3.4.1 General.- Marking shall conform to Specification MIL-M-13231.

3.4.2 Visibility.- Wherever practicable, parts and assemblies shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.4.3 Serial numbers.- The equipment shall be serial numbered.

3.5 Construction.- The equipment shall be constructed in accordance with the requirements of this specification and shall be capable of meeting the tests of Section 4.

3.6 Preproduction samples.- The contractor shall furnish preproduction samples for approval, as required by the invitation for bids and contract.

3.7 Mechanical requirements.- Printed circuit techniques, transistors, and miniature components shall be used. The equipment shall be compact and easy to service. Unitized construction shall be used to facilitate maintenance, parts stacking, and manufacture.

3.7.1 Panel mounted controls.- The following controls shall be mounted on the front panel:

(a) Order wire push button ringing switch

(b) Order wire "send" switch

(c) "Alarm-Silent" toggle switch

(d) Power "ON-OFF" switch

(e) 1KC level control adjustment

(f) Meter equalizer and test push button switch

(g) Meter selector switch

(h) Meter

(i) Transmit level toggle switch, bottom flange

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3.7.2 Channel subassembly controls.- Each individual channel subassembly shall contain a "send-measure" switch and a "receive-gain" control, mounted on the front of the assembly.

3.7.3 Line equipment subassembly controls.-

(a) Special service switch, accessible from top of assembly.

(b) Regulator toggle switch, mounted on front of assembly.

(c) Flat gain control switch, mounted on front of assembly.

(d) Equalizer control switch, mounted on front of assembly.

3.7.4 Carrier supply and order wire subassembly controls.-

(a) 4KC pilot level adjustment, accessible from top of assembly.

(b) Pilot synchronization switch, accessible from top of assembly.

3.7.5 Power supply subassembly.- A "115-230V" switch shall be provided, accessible from the top of assembly.

3.7.6 Connectors.- Provision shall be made on the front panel for the following connections:

(a) The channel drop connector shall be Cannon Type K02-10-20SN, or equal.

(b) Spiral-4 cable connections shall be made by means of a Cannon Type AN3102A-16S-1S connector, or equal, paralleled with the spiral-4 Binding Posts U-106( )/U.

(c) The order wire headset connector shall be Amphenol Type 164-7J, or equal.

3.7.6.1 A 10 foot extension cable with a Cannon Type K03-21-30PN connector, or equal, at each end shall be provided, and shall be stored in the cover of the case.

3.7.7 Lighting arrestors.- Lighting arrestors shall be 500V peak breakdown, carbon block type, Western Electric Co. No. GA-51572, or equal. They shall be mounted behind the front panel and under the chasis. Adequate low voltage protection shall be provided to protect the transistors.

3.7.8 Binding posts.- Binding Posts U-106( )/U, in accordance with Signal Corps Drawing SC-DL-72398 and including the rubber boot, shall be mounted on the front panel for connection to the spiral-4 cable and ground.

3.7.9 Fuses.- The equipment shall be adequately protected by fuses located on the front panel.

3.7.10 Indicating lamps.- Indicating lamps for special services, power, order wire signaling, and alarm circuits shall be located on the front panel.

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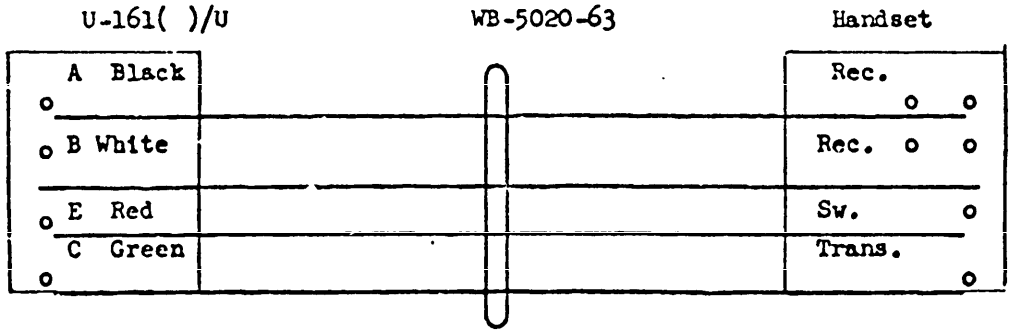
3.7.11 Power cord.- The power cord shall emerge from the front panel.

3.7.12 Chassis.- The chassis shall consist of four channel subassemblies, a line equipment subassembly, a carrier supply and order wire subassembly, and a power supply. The maximum overall depth from front to rear, including all projections, shall be 17 inches. The chassis width shall be 17.25 ± .25 inches. The front panel height shall be 5 5/32 ± 1/16 inches.

3.7.12.1 Handles and slides.- Handles shall be provided on the front panel to permit withdrawal of the chassis from the transit case. Roller slides shall be provided on both the chassis and transit case to permit easy withdrawal of the chassis from the case.

3.7.13 Transit case.- The equipment shall be housed in a fiber glass transit case. Spring loaded catches shall be provided for fastening the cover to the transit case. Handles shall be provided and shall be located so that the equipment is in balance when lifted. An interlock feature shall be provided to permit stacking of the cases.

3.7.14 Handset H-156( )/U.- Handset H-156( )/U shall be the same as Handset H-60( )/U covered by Specification MIL-H-11996(SigC) except that the cordage shall be Whitney Blake, part number 5020-63, Specification KK5020R, or equal, with a U-161( )/U Connector on one end and wired as follows:



3.8 Electrical Requirements.-

3.8.1 Transmission.-

3.8.1.1 Channel response.- The frequency attenuation characteristics of each channel shall meet the following requirements;

Frequency, cps	Output, dbm
1000	+1 = A
300	A - 1.5 ± 1.5db
3100	A ± 1db
3300	A - 1 ± 1.5db
3500	A - 2 ± 1.5db

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3.8.1.2 Channel limiting.- Channel limiting shall be as follows:

Input, dbm	Compression, db
- 14	None
- 4	0.4 or less
+ 11	7 or more

3.8.1.3 Carrier channel frequency distortion.- Any of the harmonics or demodulation components which may be produced by transmission through a carrier channel shall be at least 30 db below the fundamental of the transmitted frequency.

3.8.1.4 Carrier leak.- Carrier leaks shall be suppressed to at least 30 db below the transmission level.

3.8.1.5 4KC pilot.- The level of the 4KC pilot transmitted to the line shall be -25dbm ± 0.5 db.

3.8.1.6 Intelligible crosstalk due to channel loading.- The "near end" and "far end" crosstalk shall be not more than 34 dba.

3.8.1.7 Crosstalk due to signal tones.- Reproduction of the inserted signals in the frequency range of 300-3500 cps and any products thereof shall be in compliance with the following:

Inserted tone (-4dbm)	Levels at 4-wire receive termination	
300 cps	Fundamental down	59db
	Any other product	50db
1000 cps	Fundamental down	59db
	Any other product	50db
3500 cps	Fundamental down	59db
	Any other product	50db

3.8.1.8 Echo attenuation.- The echo attenuation shall be at least 30db.

3.8.1.9 Carrier channel noise.- Carrier channel noise shall not exceed 24 dba measured at the 4-wire receive termination.

3.8.1.10 Regulation.- Automatic regulating circuits shall maintain carrier channel net loss to within ±2db with a ±5db change in line attenuation.

3.8.1.11 Metering circuitry.- Transmission measuring equipment shall be provided to permit an overall system line-up of channel net gain to 5 ± .5db. A 1000 cps test tone shall be part of the measuring equipment.

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3.8.1.12 Special services.- The frequency attenuation characteristics of the special services shall meet the following requirements:

Frequency, cps	Output, dbm
10,000	A = 0
15,000	A ± 1db
19,000	A ± 1db
5,000	A ± 1db

3.8.1.13 Channel phase delay distortion.- The maximum delay distortion introduced by any channel shall not exceed the following requirements:

Frequency, cps	Delay, microseconds
2000	A Reference
1000	A ∠ 150
1400	A ∠ 66.7
2600	A ∠ 66.7
3000	A ∠ 150

3.8.1.14 Order wire response.- The frequency attenuation characteristics of the order wire shall meet the following requirements.

Frequency, cps	Output, dbm
1000	A
1800	A - 3 ∠ 1 db
300	A - 5 ∠ 1 db

3.8.1.15 Order wire signaling.- The 1600 cps order wire signal shall be transmitted to the line at a power of 0 dbm ∠ 1 db.

3.8.2 Crystal stability.- The frequency of the crystal oscillator circuit shall be 4000 ∠.4 cps.

3.8.3 Carrier channel impedance.- The impedance of each carrier channel shall be 600 ohms ∠10% from .3 to 3.5 kc.

3.8.4 Longitudinal balance.- The longitudinal balance of the carrier line and carrier channel impedances shall be at least 40 db.

3.8.5 Power supply.- The output voltage of the regulated power supply shall be 48 volts ∠1 volt when operated from 115 volt ∠10 percent or 230 volt ∠10 percent, 50 to 60 cycles ∠3 cycles, single phase, a.c. power source.

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3.9 4-Channel hybrid transformer unit.- The 4-channel hybrid transformer unit shall consist of a chassis and a front panel. The width of the unit shall be the same as that of the Telephone Terminal AN/TCC-24( ). The depth shall not exceed 6 inches. The height of the front panel shall be either 1-21/32 ±1/16 inches or 3-13/32 ±1/16 inches. A dust cover shall be provided.

3.9.1 Connectors.- The 4-wire channel drop connector shall be mounted on the front panel and shall be Cannon type KO2-21-30SN, or equal. Binding Posts U-106( )/U, in accordance with Signal Corps Drawing SC-DL-72398 and including rubber boots, shall be mounted on the front panel for the 2-wire and 4-wire channel drop connections, special services, order wire extension circuit, and grounds.

3.9.2 Hybrid transmission loss.- The hybrid transmission loss shall not exceed 4db.

3.9.3 Trans-hybrid loss.- The trans-hybrid loss shall be more than 30 db.

3.9.4 Each channel Hybrid shall be provided with a 2-wire-4-wire option switch.

3.10 Service conditions.- The equipment shall meet the following service conditions:

3.10.1 Equipment nonoperating.- The equipment shall comply with the operational requirements of 3.10.2 after subjection to any of the following nonoperating conditions successively or in combinations encountered during worldwide short-term storage and transit.

(a) Temperature.- Exposure in the range of ±160°F to -80°F; however, exposure at the high temperature extreme not to exceed 4 hours and at the low temperature extreme not to exceed 24 hours at any one time.

(b) Relative humidity. Relative humidity up to 97 percent, for an indefinite period of time, and exposure at 100 percent relative humidity, with condensation for 4 hours.

(c) Elevation. Elevation up to 25,000 feet above sea level.

(d) Rain. Two inches of rainfall per hour, with wind at nominal 20 miles per hour.

(e) Vibration, bounce, and shock. As described in 3.11.

3.10.2 Equipment operating.- The equipment shall meet the performance requirements of this specification while subjected to any of the following conditions successively or in combinations likely to be encountered during worldwide operations.

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(a) Temperature. Ambient temperature in the range of +132°F to -40°F. Exposure at the high temperature extreme not to exceed 4 hours and at the low temperature extreme not to exceed 72 hours.

(b) Relative humidity. Relative humidity up to 97 percent, and exposure at 100 percent relative humidity with condensation for 4 hours.

(c) Elevation. Elevation up to 10,000 feet above sea level.

3.11 Vibration, bounce and shock.

3.11.1 Vibration. Except for internal resonance of specified parts and sub-assemblies, the equipment shall have no mechanical resonance below 55 cycles per second, when shock mounts, if any, are blocked or removed.

3.11.2 Bounce. The equipment shall meet full specification performance and show no physical damage except surface abrasion after subjection to the test of 4.12.6.

3.11.3 Shock, bench handling. The equipment shall meet full specification performance and show no physical damage after subjection to the test of 4.12.7.

3.11.4 Shock, drop. The equipment shall be operable after subjection to the test of 4.12.8. Minor mechanical damage is permissible.

3.12 Radio interference suppression. The equipment shall conform to the requirements of Military Specification MIL-I-11748 for Class I equipments.

3.13 Interchangeability. Corresponding components, replaceable sub-assemblies,,and replaceable parts on contract shall be physically and functionally interchangeable, as units without modification thereof or of other items with which the units are used. When dimensions, ratings, characteristics, etc., are not specified, the manufacturer's design limits shall be used to determine compliance with the foregoing. If the contractor is in doubt as to whether a particular sub-assembly or part is to be considered replaceable, the contracting officer shall be consulted.

3.14 Preconditioning. The equipment shall be capable of meeting the inspection of Section 4, without subsequent processing, after subjection to the bounce preconditioning of 4.4.

3.15 Technical literature and running spare parts. Technical literature and running spare parts shall be furnished as specified by the contract. Running spare parts shall be identical to corresponding parts in the equipment furnished on the order.

3.16 Workmanship. Workmanship on components covered by this specification shall be such as to meet all the requirements of this specification and any referenced subsidiary specification, drawing, or other document, when inspected in accordance with 4.15, including the applicable portions of the following paragraphs:

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(a) In Specification MIL-P-11268:

- General requirements for plastic material and plastic parts.
- Wiring and cabling, including:
  - Slack.
  - Protection
  - Clearance
  - Splicing
  - Connections, general
  - Grounding, general
  - Shielding on wire and cable
- Flux and cleaning agents for soldering.
- Process for soldering.
- Cleaning of equipment.
- Riveting.
- General requirements for securing of parts.
- Cleaning of welding.
- Process for welding.
- Controls.

(b) In this Specification:

- Marking
- Finish
- Connectors
- Binding Posts
- Power cord

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4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of inspection. Inspection shall be classified as follows:

(a) Preproduction inspection (does not include preparation for delivery). (See 4.2.)

(b) Acceptance inspection of items before they are prepared for delivery. (See 4.3.)

(c) Acceptance inspection of preparation for delivery. (See 4.16 and Section 5.)

4.2 Preproduction inspection.

4.2.1 Preproduction inspection of equipment. This inspection shall consist of the group A inspection and nondestructive group B inspection specified in tables II, and III respectively, and the preproduction inspection specified in table I. Other nondestructive inspection on the preproduction samples may be performed to determine compliance with specified requirements. The preproduction inspection will normally be performed in this order: (1) vibration, (2) bounce, (3) shock, benchhandling, (4) shock, drop. Other preproduction inspection may precede, follow, or be interspersed between the foregoing.

Table I.- Preproduction inspection

Inspection	Rqt. Par.	Insp. Par.
Temperature	3.10	4.12.1
Moisture resistance	3.10	4.12.2
Altitude	3.10	4.12.3
Rain	3.10.1	4.12.4
Vibration	3.11.1	4.12.5
Bounce	3.11.2	4.12.6
Shock, benchhandling	3.11.3	4.12.7
Shock, drop	3.11.4	4.12.8
Radio interference suppression	3.12	4.13

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4.3 Acceptance inspection. This inspection shall include group A and group B, inspection as specified in 4.3.1 through 4.3.2.3. Group B inspection shall normally be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection. However, the order may be varied when the Government considers it more practical to select separate samples for group B inspection. Each unit which will be subjected to group A or group B, inspection shall be preconditioned after final assembly. (See 3.14)

4.3.1 Group A inspection. This inspection (including sampling) shall conform to table II and Standard MIL-STD-105. It shall be performed in any order which is satisfactory to the Government inspector.

Table II - Group A inspection

Inspection	Reqt Par.	Insp. Par.	AQL Major	AQL Minor
Visual and mechanical				
Channels 1 through 4 subassemblies			2.5 dphu	10.0 dphu
Line Equipment subassembly	3.7.12 and 3.16	4.15	2.5 dphu	10.0 dphu
Carrier supply and order wire subassembly			4.0 dphu	15.0 dphu
Power supply subassembly			2.5 dphu	10.0 dphu
Complete assembly	3.7.12	4.15	1.5 dphu	6.5 dphu
Hybrid transformer	3.9	4.15	1.0 dphu	4.0 dphu
Electrical				
Transmission				
Channel response	3.8.1.1	4.5.2		
Channel limiting	3.8.1.2	4.5.3	4.0 dphu for the combined *	
Regulation	3.8.1.10	4.5.4		
Metering circuit	3.8.1.11	4.5.5		

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Inspection	Rqt. Par.	Insp. Par.	M AQL Major	Minor
Channel distortion	3.8.1.3	4.5.10		
4KC pilot	3.8.1.5	4.5.12		
Crystal stability	3.8.2	4.6		
Longitudinal balance	3.8.4	4.8		
Power supply	3.8.5	4.9		
Hybrid transformer				
Hybrid transmission loss	3.9.2	4.10	1.5 dphu combined	*
Trans-hybrid loss	3.9.3	4.11		
Operational				
Order wire response	3.8.1.14	4.5.8		
Order wire signaling	3.8.1.15	4.5.9		
Intelligible crosstalk due due to channel loading	3.8.1.6	4.5.13	1.0% combined	*
Crosstalk due to signaling tones	3.8.1.7	4.5.14		
Carrier channel noise	3.8.1.9	4.5.16		

* All electrical defects are of major category.

4.3.2 Group B inspection. This inspection (including sampling) shall conform to table III and the Appendix to Standard MIL-STD-105. Unless otherwise specified herein, normal inspection shall be used at the start of the contract. The reduced inspection procedure shall be R-1. Disposition of rejected product (sample units and inspection lots) shall be in accordance with Standard MIL-STD-105 and 4.3.4.

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4.3.2.1 Group B sampling plans. The group B sampling plans, as listed in Table III, shall be as follows:

Group B plan	AQL	Inspection level for Normal Inspection	Inspection level for Reduced Inspection
B-1	4.0%	L-8	L-6
B-2	6.5%	L-7	L-5
B-3	10.0%	L-6	L-4
B-4	15.0%	L-5	L-3

4.3.2.2 Order of inspection within Group B. Group B inspection shall be performed in any order which is satisfactory to the Government inspector.

4.3.2.3 Procedure in case of failure. When an inspection lot is rejected, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken. If the contractor and Government inspector cannot agree on the effectiveness of the corrective action, the matter shall be referred to the contracting officer for resolution.

Table III - Group B Inspection

Inspection	Rqt. Par.	Insp. Par.	Sampling Plan
Transmission			
Special services	3.8.1.12	4.5.6	
Delay distortion	3.8.1.13	4.5.7	
Carrier leak	3.8.1.4	4.5.11	B-2
Echo Attenuation	3.8.1.8	4.5.15	combined
Output impedance	3.8.3	4.7.1	
Input impedance	3.8.3	4.7.2	
Interchangeability inspection	3.13	4.14	

4.3.3 Reinspection of conforming Group B, sample units. Unless otherwise specified, sample units which have been subjected to and passed Group B inspection, may be accepted on contract, provided that they are resubjected to and pass Group B inspection after repair of all visible damage.

4.3.4 Disposition of nonconforming product. When defective sample units or rejected inspection lots are resubmitted for acceptance, such product shall be suitably tagged or identified by equivalent means to indicated the cause of failure and means employed to correct the fault. The record shall be presented to the Government when the product is resubmitted and shall become the property of the Government.

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4.4 Bounce preconditioning. The equipment, with shock mounts (if any) removed or blocked, shall be placed in its normal operating position on the table of the Package Tester as made by the L.A.B Corporation, Skaneatles, N.Y., or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32-inch thick strip of material under one corner or edge of the equipment to a distance of three inches as the equipment bounces. The equipment shall be subjected to this preconditioning for 1 minute. After bounce preconditioning, the equipment shall not be repaired, alined, cleaned, or otherwise changed prior to subjection to acceptance inspection.

4.5 Transmission tests.

4.5.1 Test set-up. The equipment under test shall be arranged as shown in Figure 1 and lined up in both directions to provide a channel net gain of 5 db, using the internal metering circuit and oscillator. The "local-remote" switch shall be in the local position for one terminal and in the remote position for the other terminal. This arrangement shall be used to make the following measurements.

4.5.2 Channel response. Connect a Signal Generator SG-15( )/PCM adjusted for an output of -4 dbm to the channel 4-wire transmit pair of one terminal. With a Decibel Meter ME-22( )/PCM set for 600 ohm termination connected to the channel 4-wire receive pair, measure the frequency-attenuatuin characteristics. Make measurements on all channels in both directions for compliance with 3.8.1.1.

4.5.3 Channel limiting. Using the same arrangement as indicated in 4.5.2, adjust the oscillator to 1000 cps and measure the channel net loss with the oscillator set to -11 dbm, -4 dbm and +11 dbm. Make measurements on channels 1 and 4 in each direction for compliance with 3.8.1.2.

4.5.4 Regulation. Using the same arrangements as indicated in 4.5.2, adjust the oscillator to 1000 cps and measure the channel net loss. Vary the line attenuation ±5 db and measure the channel net loss. Make measurements on channel 1 and 4 in each direction for compliance with 3.8.1.10.

4.5.5 Metering circuit. Using the same arrangement as indicated in 4.5.2, measure the channel net loss at 1000 cps. Make measurements on channels 1 and 4 in each direction for compliance with 3.8.1.11.

4.5.6 Special services. Connect a Signal Generator SG-15( )/PCM adjusted for an output of 0 dbm to the Special Service transmit pair. Measure the frequency-attenuation characteristics with a Decibel Meter ME-22( )/PCM connected to the special service receive pair of the other terminal. Make measurement in each direction for compliance with 3.8.1.12.

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4.5.7 Delay distortion. See Note 6.4.

4.5.8 Order-wire response. Connect a Signal Generator SG-15( )/PCM adjusted for an output of 0 dbm to order-wire 2-wire extension of one terminal. With a Decibel Meter ME-22( )/PCM set for 600 ohm termination connected to the order-wire extension termination of the other terminal, measure the frequency attenuation characteristics. Make measurements at 300, 1000, and 1800 cps in both directions for compliance with 3.8.1.14.

4.5.9 Order-wire signaling. Using the same arrangement as in 4.5.8, measure the 1600 cps level while depressing the order wire push button, to determine compliance with 3.8.1.15. Repeat measurements for the other terminal.

4.5.10 Channel distortion. Connect a Signal Generator SG-15( )/PCM adjusted for an output of -4 dbm to the channel 4-wire transmit pair. Connect a Wave Analyzer having a 600 ohm termination (internal or external) to the channel 4-wire receive pair. Make measurements at 300 and 1000 cps in all channels in each direction for compliance with 3.8.1.3.

4.5.11 Carrier leak. Connect a Wave Analyzer having a 600 ohm termination (internal or external) to the channel 4-wire receive pair. Measure the carrier leaks as follows: 8 kc in channel 1, 12 kc in channel 2, 16 kc in channel 3, and 20 kc in channel 4. Make measurements on each terminal for compliance with 3.8.1.4.

4.5.12 4 kc pilot. Connect a high impedance wave analyzer across the transmit pair of the line side and measure the 4 kc output level with the transmit level toggle switch in the 0 position. Repeat measurement for the other terminal. Pilot level shall be in compliance with 3.8.1.5.

4.5.13 Intelligible cross talk due to channel loading. With 3 channels each loaded with a -6 VU talker using a TA-312/PT Telephone Set at the channel 4-wire transmit termination, intelligible crosstalk introduced by the system into the remaining idle channel of both terminals, as measured with a Transmission Measuring Set TS-559( )/PT using F1A weighting at the channel 4-wire receive termination, shall be in comformance with 3.8.1.6. Make measurements loading channels 1-2-3, 2-3-4, 3-4-1 and 1-2-4.

4.5.14 Crosstalk due to signal tones. Connect a Signal Generator SG-15( )/PCM adjusted to -4dbm to the channel 4-wire transmit pair. With a Harmonic Analyzer terminated in 600 ohms connected to the channel 4-wire receive pair of the other 3 idle channels on both terminals, measure the fundamental and other products for compliance with paragraph 3.8.1.7. Apply tones to each channel in turn and repeat measurements. Make measurements in both directions at 300, 1000, and 3500 cps.

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4.5.15 Echo attenuation. Connect a Signal Generator SG-15 adjusted to -4 dbm to the channel 4-wire transmit pair. Connect a Decible Meter ME-22( )/PCM in the 600 ohm terminating position to the channel 4-wire receive pair of the same channel and terminal. Make measurements at 300 cps, 1000 cps, and 3500 cps on all channels for compliance with 3.8.1.8.

4.5.16 Carrier channel noise. Connect a Transmission Measuring Set TS-559( )/FT adjusted for FIA line weighting to the channel 4-wire receive pair. Make measurements on all channels in each terminal for compliance with 3.8.1.9.

4.6 Crystal stability. Connect a frequency counter across the output terminals of the 4 kc oscillator circuit. Make measurements on each terminal with the syncronization switch in local position for compliance with 3.8.2.

4.7 Carrier channel impedance.

4.7.1 Output impedance. Connect on oscilator having an output of -4 dbm to the channel 4-wire transmit pair at one terminal and an electronic Multimeter ME-30( )/U to the channel 4-wire receive pair of a corresponding channel of the other terminal. Measure the voltage. Connect a variable resistance box across the voltmeter and 4-wire receive pair. Vary resistance until meter reads 1/2 original value and record. Impedance (resistance) shall be in conformance with 3.8.3. Make measurements on all channels of each terminal at 300 cps, 1000 cps, and 3500 cps.

4.7.2 Input impedance. Connect an electronic Multimeter ME-30( )/U and an oscillator having an output of -4 dbm to the channel 4-wire transmit pair. Measure the voltage. Substitute a variable resistance box for the channel 4-wire transmit pair. Vary the resistance until the voltage across the resistance box corresponds to that across the channel 4-wire transmit pair. Impedance (resistance) shall be in conformance with 3.8.3. Make measurements on all channels of each terminal at 300 cps, 1000 cps, and 3500 cps.

4.8 Longitudinal balance.

a. Channel 4-wire transmit pair. Using the arrangement shown in Figure 2A, make measurements at 300 cps, 1000 cps, and 3500 cps on all channels of each terminal for compliance with 3.8.4.

b. Channel 4-wire receive pair. Using the arrangement shown in Figure 2b, make measurements at 300 cps, 1000 cps, and 3500 cps on all channels of each terminal for compliance with 3.8.4.

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4.9 Power supply. With an Electronic Multimeter TS-505( )/U connected across the output terminals of the regulated power supply, make measurements on each terminal for compliance with 3.8.5.

4.10 Hybrid transmission loss. Connect a Signal Generator SG-15( )/PCM adjusted for an output of 0 dbm to the 2-wire side of the hybrid. Terminate the 4-wire receive pair with a 600 ohm resistor. With a decibel Meter ME-22 in the terminating position, connected to the 4-wire transmit pair, make measurements at 300 cps, 1000 cps, and 3500 cps. Interchange the decibel meter and 600 ohm resistor and repeat measurements. Make measurements on all hybrids for compliance with 3.9.2.

4.11 Trans-hybrid loss. Connect a Signal Generator SG-15( )/PCM adjusted for an output of 0 dbm to the 4-wire receive pair. Terminate the 2-wire side with 600 ohms in series with 2 micro-farads. Connect an electronic Multimeter ME-30( )/U, terminated in 600 ohms to the 4-wire transmit pair. Make measurements on all hybrids at 300 cps, 1000 cps, and 3500 cps for compliance with 3.9.3.

4.12 Service condition tests. Telephone Terminal AN/TCC-24( ) shall be subjected to all of the service tests. The hybrid transformer rack shall not be subjected to the rain, bounce or drop tests.

4.12.1 Temperature. The equipment shall be subjected to the temperature cycle shown on MIL-STD-169. Tests shall be made at stop 4 and at step 8 as well as steps required by MIL-STD-169.

4.12.2 Moisture-resistance test.

4.12.2.1 Test conditions.

(a) Do not remove equipment from the humidity chamber for measurements.

(b) Start measurements not more than 5 minutes after power is applied to the equipment. Complete measurements as rapidly as possible. Do not leave power on after measurements have been completed.

(c) Test sealed equipment with the seal intact, except during the last two cycles. Bring out leads for electrical measurements through a hole in the equipment enclosure and seal against entrance of moisture. Perform the last two cycles, with the chassis removed from its enclosure.

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4.12.2.2 Test procedure. The sample equipments will be tested as follows:

(a) Dry at 130 ±5°F for 24 hours.

(b) Condition at 77° ±5°F and 40 to 50 percent relative humidity for 24 hours.

(c) Take measurements as specified in 4.12.2.3.

(d) Subject to continuous cycling for a total of five 48-hour cycles. Temperature, relative humidity, and period of time for each portion of the cycle shall conform to MIL-STD-170. The measurements specified in 4.12.2.3. shall be made during periods shown on the drawing.

(e) After cycling has been completed, condition the equipments for 24 hours at 77 ±5°F at 40 to 60 percent relative humidity. No repair or replacement of parts shall be made. After adjustment, the equipments shall meet full specification performance for those measurements specified in 4.12.2.3.

4.12.2.3 Performance. Measurements made during test and specified performar during cycling shall conform to the electrical requirements of Section 3.

4.12.2.4 Failure. If any component fails to meet the performance specifieu 4.12.2.3 during cycling, it shall be realined or readjusted once. If the component then fails to meet 4.12.2.3 or fails subsequently during cycling, it does not pass the test In addition, if the component fails to meet full specification requirements after conditioning and adjustment, as specified in 4.12.2.2.(e) it does not pass the test.

4.12.3 Altitude test. The equipment shall be placed in an altitude chamber at normal conditions of temperature, pressure, and humidity, and shall be tested to determine compliance with electrical requirements. The pressure shall then be reduced to 20 inches of mercury and stabilized for 2 hours. The equipment shall again be tested. The pressure shall then be lowered to 11 inches of mercury and the chamber maintained at this pressure for another 2-hour period. The pressure shall then be raised to 29.9 inches of mercury and tested again. There shall be no degradation of performance in the second and third measurements from those established in the first test.

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4.12.4 Rain test. The transit case, with the equipment in it shall be tested, step by step, as follows:

4.12.4.1 Dry at 150 ±5 F for 48 hours.

4.12.4.2 Condition for 4 hours at 77 ±5F and 40 to 60 percent relative humidity.

4.12.4.3 Place the case, in its normal operating position, in a test chamber maintained at 77 ±5 F. All simulated rainfall to fall on exposed surfaces of the case. The water shall be at a temperature of 65 ± 5F and shall fall freely at a rate of 2 inches per hour. Maintain this condition for one hour.

4.12.4.4 With the rainfall continuing at the 2-inch rate, introduce a simulated wind blowing horizontally at 20 ±5 miles per hour against one side of the component. The water shall fall on exposed surfaces and be blown by the wind against one side of the component. Maintain this condition for one hour.

4.12.4.5 Rotate the case 90 degrees horizontally to expose a second side of the case to the wind and rain. Maintain this condition for one hour.

4.12.4.6 Continue as in 4.12.4.5 until all four sides of the case have been exposed to the wind and rain, each side for one hour.

4.12.4.7 Open the case. There shall be no visible evidence of leakage.

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4.12.5 Vibration test. The equipment shall be tested for resonant frequencies below 45 cycles per second, to determine compliance with 3.11.1. as follows:

a. The equipment shall be fastened in its normal mounting position on a vibration table that can be controlled within 10 percent of the specified amplitude. Equipment shock mounts, if any, shall be blocked. The vibration table shall provide approximately sinusoidal vibration.

b. The equipment shall be vibrated successively in three mutually perpendicular directions that are parallel respectively to the edges of the equipment, over a frequency range of 10 to 55 cycles per second, in 1-cycle-per-second steps. The total excursion shall be constant at 1/64 inch.

c. Mechanical resonance, if any, of the complete structure, of subassemblies, and of component parts shall be determined visually by means of a strobotac, as made by the General Radio Corporation, Cambridge, Massachusetts, or equal, or by energizing the equipment and detecting mechanical vibration through electrical output indications.

d. If it is not practical to determine completely any mechanical resonances on the assembled equipment unit because of its compactness or the inaccessibility of its subassemblies or component parts, the unit shall be broken down into logical subassemblies for the above vibration test. Such subassemblies shall be secured to the vibration table in the same manner as they are secured in the equipment unit.

4.12.6 Bounce test. The equipment in its transit chest shall be placed on the table of the package tester as made by L.A.B. Corps., Skaeneatles, N. Y., or equal, and shall be constrained from horizontal motion of more than 2 inches by suitable wooden fences. The package tester, shafts in phase, shall be operated at a speed of 285 rpm ±1%, for a total of 3 hours. During each 1/2 hour period of the test, the transit chest (which contains equipment) shall rest on a different face.

4.12.7 Shock test: bench handling. The chassis-and-front panel assembly shall be removed from its enclosure, as for servicing, and placed in a suitable position for servicing on a solid 2-inch fir bench top. The test shall be performed as follows, in a manner simulating shocks liable to occur during servicing:

a. Tilt up the assembly through an angle of 30°, using one edge of the assembly as an axis, and permit the assembly to drop back freely to the horizontal. Repeat, using other practicable edges of the same horizontal face as axes, for a total of four drops.

b. Repeat "a", with the assembly resting on other faces, until it has been dropped for a total of 4 times on each face on which the assembly might practicably be placed during servicing.

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4.12.8 Shock test: drop. The equipment, in its transit case shall be dropped a distance of 48 inches on each corner, edge, and face (total of 26 drops). Drops shall be made from a quick-release hook; and Acme Drop Tester, or equal; a Conbur Tester (10 degree inclined plan), or equal; or a basket-type pendulum tester. The floor or barrier receiving the impact shall be 2 inch fir, back by concrete or a rigid steel frame.

4.13 Radio interference suppression. The equipment shall be tested to determine compliance with 3.12.

4.14 Inspection for interchangeability. The dimensions listed below shall be gaged or measured (see Table III) to determine compliance with the physical interchangeability requirement of 3.13. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

(a) External and internal dimensions of cases, covers, and insertable subassemblies, when such dimensions affect mating of parts.

(b) Internal dimensions of cavities in cases and assemblies, when such dimensions affect acceptability of insertable subassemblies.

(c) Location of hinges and fasteners on separable parts or assemblies which must mate together, such as cases, covers, and mountings.

(d) Location of connectors, locking pins, fasteners, slides, and mountings which receive mating parts of plug-in assemblies and major components.

(e) Location of connectors, locking pins, fasteners, slides, and mountings on plug-in assemblies.

(f) Size and form af special threads.

4.15 Visual and mechanical. The equipment shall be examined for the defects listed in Standard MIL-STD-252.

4.16 Acceptance inspection of preparation for delivery.

4.16.1 Preservation and packaging. Inspection of preservation and packaging shall be as specified in Specification MIL-P-116. Classification of defects shall be as shown in Table IV.

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TABLE IV

MAJOR DEFECTS

1. Use of improper or defective material.
2. Quantity in unit package not as specified.
3. Incorrect packaging method applied.
4. Cushioning or padding omitted.
5. Cushioning inadequate for the physical and mechanical protection of the item.
6. Unsealed, punctured, or improperly sealed barrier bag, wrap, or envelope.
7. Stock number omitted, incorrect, or illegible.
8. Nomenclature omitted, incorrect, or illegible.
9. Marking of quantity of items in package omitted, incorrect, or illegible.
10. Different stock numbered items in the same unit package.

MINOR DEFECTS

1. Item not properly blocked or braced within the unit package to prevent movement.
2. Packaging material damaged.
3. Any item of marking information other than 7 thru 9 omitted, incorrect, or illegible.

4.16.2 Packing and marking of exterior containers. Packing and marking of exterior containers shall be given visual inspection for the defects listed below and to determine conformance with the approved process sheet furnished by the contractor as required by the contract. This inspection shall conform to the Appendix to Standard MIL-STD-105. Inspection level L-8 shall be used for normal inspection and L-6 for reduced inspection. Unless otherwise specified herein, normal inspection shall be used at the start of the contract. The reduced inspection procedure shall be R-1. The AQL for major defects shall be four percent and the AQL for minor defects shall be ten percent. Classification of defects shall be as shown in Table V.

TABLE V

MAJOR DEFECTS

1. Use of improper or defective material.
2. Quantity in pack not as specified.
3. Gross weitht in excess of specified amount.
4. Box closure not as specified.
5. Type, grade, class, and style of the shipping container not as specified.
6. Strapping omitted.
7. Strapping inadequate or incorrectly applied.
8. Items not adequately blocked, braced, or cushioned within the shipping container to prevent movement or damage.
9. Shipping documents or packing list omitted.
10. Stock number omitted, incorrect, or illegible.
11. Nomenclature omitted, incorrect, or illegible.
12. Marking of quantity of items in pack omitted, incorrect, or illegible.
13. Destination marking omitted, incorrect, or illegible.
14. Service designation (color marking) omitted.
15. Specified special marking and labeling such as MAP labels, shipment digit markings, etc., omitted, incorrect, or illegible.
16. Overseas code marking omitted, incorrect, or illegible.

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MINOR DEFECTS

1. Unsealed carton.
2. Defective taping or sealing of carton.
3. Any other box defect which may be considered minor by definition of Standard MIL-STD-105.
4. Any item of required marking information other than 10 thru 16 listed under major defects omitted, incorrect, or illegible.

4.16.2.1 Inspection lot. A lot for visual inspection of the pack shall be all completed packs which are identical and are submitted for inspection at one time.

4.16.2.2 Procedure in case of failure. If an inspection lot is rejected the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken. If the contractor and Government inspector cannot agree on the effectiveness of the corrective action, the matter syall be referred to the contracting officer for resolution.

4.16.2.3 Disposition of nonconforming product. Disposition of rejected lots and defective sample units shall be in accordance with Standard MIL-STD-105. When resubmitted for acceptance, such product shall be suitably tagged or identified by equivalent means to indicate the cause of failure and means employed to correct the fault. The required information shall be presented to the Government when the product is resubmitted and shall become the property of the Government.

4.17 Rough handling test (preparation for delivery). When rough handling test is required by the contract, (see 6.2(n)3) the following functional tests shall be conducted to determine freedom from operational malfunction caused by the rough handling.

Operational Tests

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level A. Telephone Terminal AN/TCC-24( ) shall be preserved, packaged and tested in accordance with the procedures specified for the designated methods as prescribed in Specification MIL-P-116 and described herein.

5.1.1.1 Technical manual. Each technical manual shall be packaged Method IC-3, as follows: Inclose each technical manual within a close-fitting, Type II, Class B bag, conforming to Specification MIL-B-117. Heat seal bag to effect closure..

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5.1.1.2 Hybrid transformer unit. Each unit shall be packaged Method IC-5, as follows: Cushion each unit on all surfaces with cells or pads or both, fabricated of fiberboard, having a minimum dry bursting strength of 175 pounds, and designed to absorb the shock of impact, encountered in handling and transit. Cells and pads shall have a water resistance equal to that of the box in which they are used. Place each cushioned unit within a close-fitting, Type I, Class 2, Grade 6, box conforming to Specification PPP-B-636. Seal the box as specified in the appendix of the box specification and in addition seal all joints and seams with waterproof, pressure sensitive tape, Type 3, Class 3, conforming to Specification PPP-T-60.

5.1.1.3 Accessories. Stow the accessories within the case cover and secure with fastenings provided.

5.1.1.4 Telephone terminal. Each terminal shall be packaged Method IC-5 as follows: Position case cover with stowed accessories over the terminal and secure all fastenings. Continue packaging, following procedures specified in 5.1.1.2.

5.1.2 Level C. Telephone Terminal AN/TCC-24 shall be packaged in accordance with commercial practice and in a manner that will afford protection against corrosion, deterioration and physical damage, during direct shipment to the first receiving activity.

5.2 Packing.

5.2.1 Level A. Components of Telephone Terminal AN/TCC-24 shall be packed together within a Class 2, Style 4, nailed, wood box, conforming to Specification PPP-B-621. Fill voids with cells or pads or both, fabricated of fiberboard, conforming to Table II of Specification PPP-B-636 to prevent movement or damage in handling and transit. Fabricate the box to fit the contents snugly. Place the packaged technical manual, between the contents and the list of the shipping container. Box closure shall be in accordance with the box specification.

5.2.1.1 Metal strapping. Shipping containers shall be strapped, with flat steel strapping, in conformance with the requirements of the appendix of the container specification, only for direct shipment to ports.

5.2.2 Level B. Telephone Terminal AN/TCC-24 shall be packed as specified in 5.2.1 except that the box shall conform to Class I, Style 4, of Specification PPP-B-621.

5.2.3 Level C. Telephone Terminal AN/TCC-24 shall be packed for shipment in a manner conforming to the requirements of Uniform Freight Classification for railshipment, National Motor Freight Classification for truck shipment, Parcel Post Regulations and the regulations of other carriers as applicable to the mode of transportation employed, at the lowest transportation rate.

5.3 Marking. Interior packages and exterior shipping containers shall be marked in accordance with the applicable provisions of Standard MIL-STD-129.

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6. NOTES

6.1 Intended use. Telephone Terminal AN/TCC-24( ), Transistorized provides 1 + 4 channel operation over loaded spiral-4 cable and over radio relay equipment.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification and any amendment thereto.

b. Level of packaging and level of packing required for shipment.

c. Preproduction samples required:

2 complete Telephone Terminal AN/TCC-24( )
2 complete hybrid transformer units.

d. Marking and shipping of samples

e. Place of final inspection.

f. Technical literature required.

g. Quantity of running spare parts.

h. Preproduction pack(s) as follows:

(1) Makeup of pack(s)
(2) Number of each kind of pack to be submitted
(3) Inspection that will be performed thereon-including rough handling test, which will not be performed as acceptance inspection.
(4) Five copies of process sheet, using format shown on Drawing SC-D-33073 and accompanied by such drawings, sketches, and other data as may be necessary to completely describe the procedure followed in fabrication of the preproduction pack(s) and to identify the items therein. These data should be submitted to the contracting officer with the preproduction pack.
(5) A packaging technician from the procuring agency will evaluate the preproduction pack(s).

i. The specific paragraphs of section 5 which are applicable to the particular procurement.

6.3 Nomenclature. The parenthesis in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: AN/TCC-24W. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

6.4 Delay distortion test. The contractor shall devise the method for making the delay distortion test of 4.5.7 and submit the procedure to the contracting officer for his approval.

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6.5 Definition of level B. Level B packaging requirements detailed herein are designed to furnish protection against corrosion, deterioration, and physical damage during domestic shipments; covered storage for a period of three years or less; and distribution within the continental limits of the United States.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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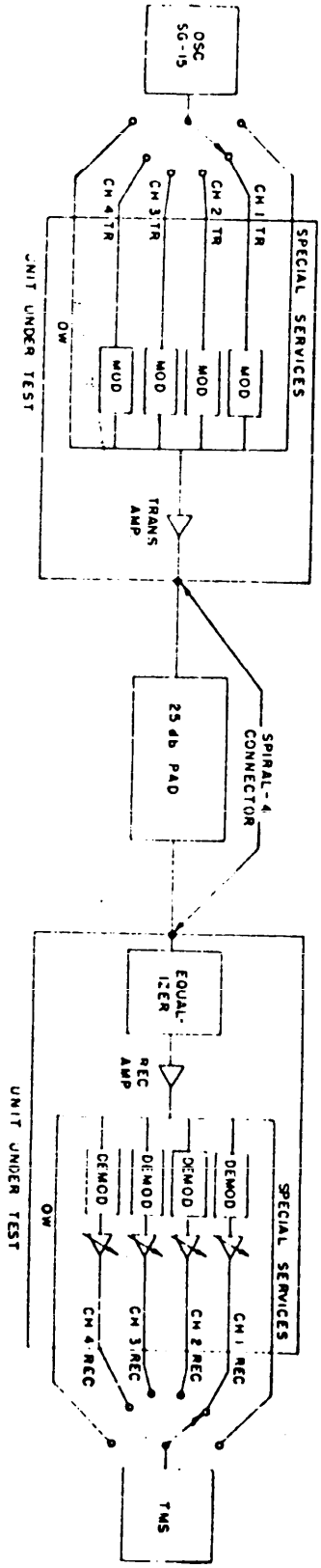


FIG 1 TRANSMISSION MEASUREMENTS

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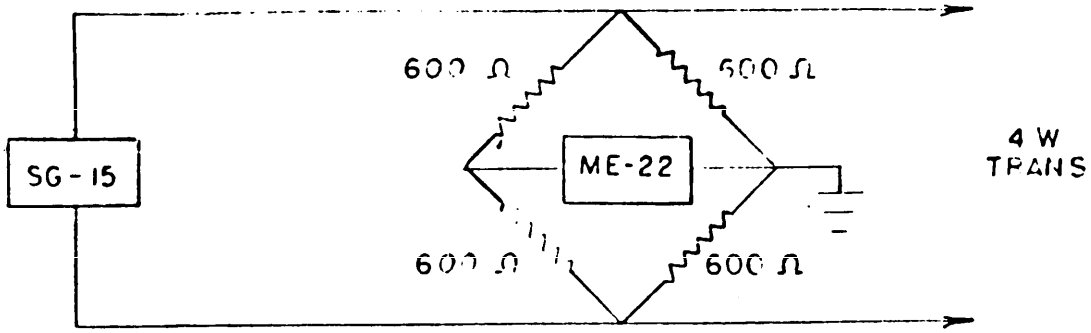


FIG 2A

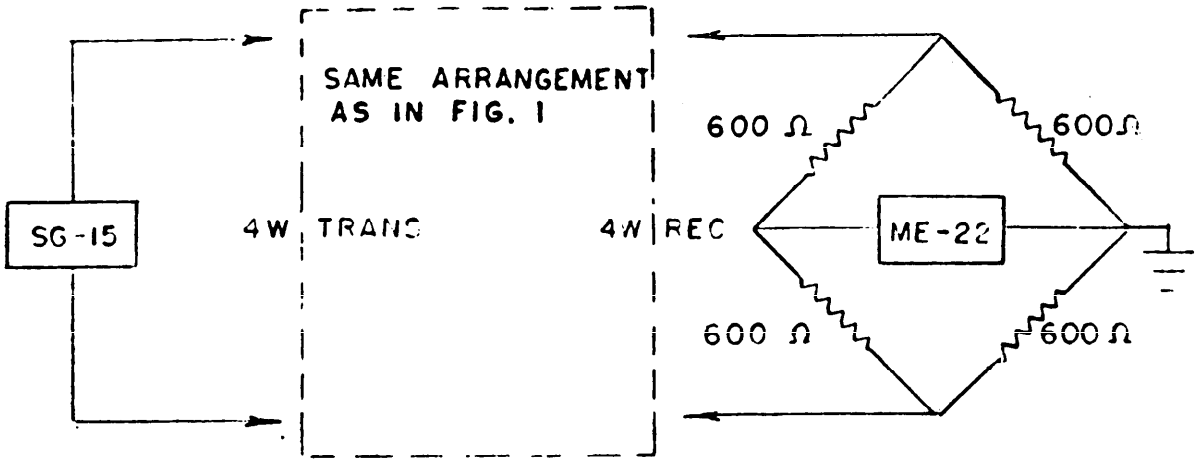


FIG 2B

ALL RESISTORS 600 Ω ±.1%

LONGITUDINAL BALANCE

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

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DEPARTMENT OF THE NAVY

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

COMMANDING GENERAL
U. S. ARMY ELECTRONICS COMMAND
ATTN: AMSEL-PP-PTS
FORT MONMOUTH, N.J. 07703

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